Vector Analysis

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- Class group ID? – send a test email
- CR? – send a WhatsApp message

The dot product

- Given two vectors **A** and **B**, the *dot product*, or *scalar product*, is defines as the product of the magnitude of **A**, the magnitude of **B**, and the cosine of the smaller angle between them:

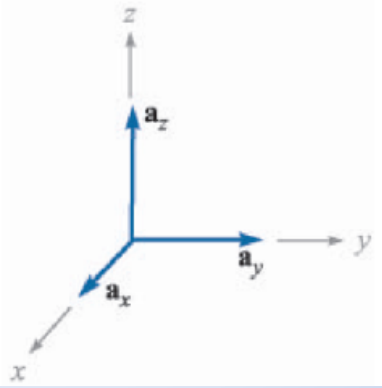
$$\mathbf{A}\cdot\mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta_{AB}$$

- The dot product is a scalar, and it obeys the commutative law:

$$\mathbf{A}\cdot\mathbf{B} = \mathbf{B}\cdot\mathbf{A}$$

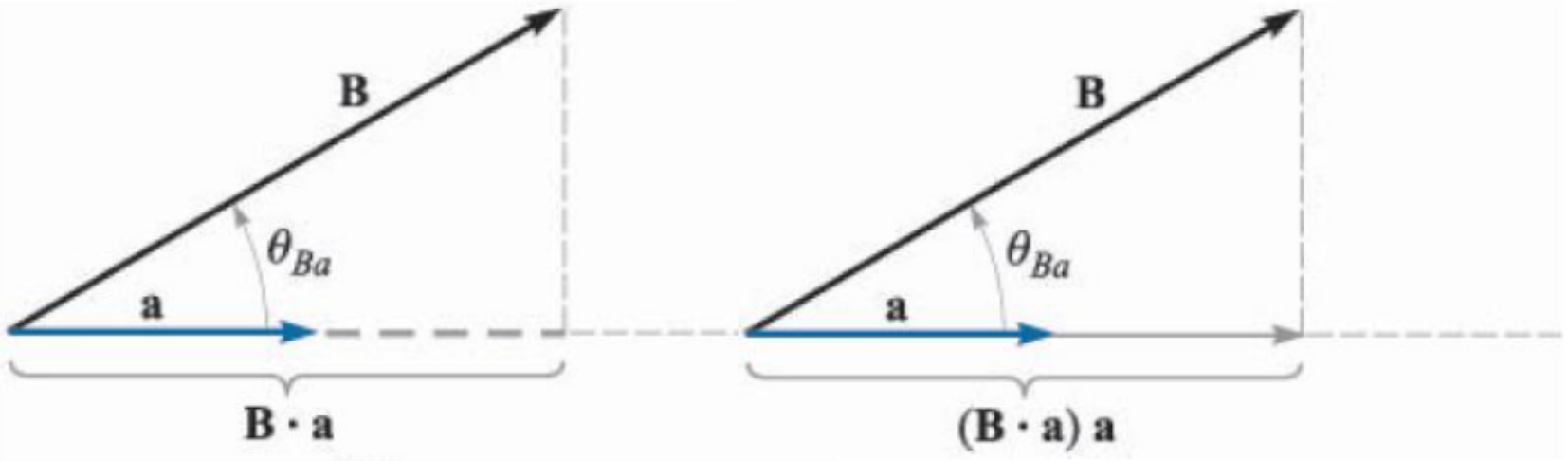
- For any vector $\mathbf{A} = A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z$ and $\mathbf{B} = B_x\mathbf{a}_x + B_y\mathbf{a}_y + B_z\mathbf{a}_z$,

$$\mathbf{A}\cdot\mathbf{B} = A_xB_x + A_yB_y + A_zB_z$$



The dot product

- One of the most important applications of the dot product is that of finding the component of a vector in a given direction.



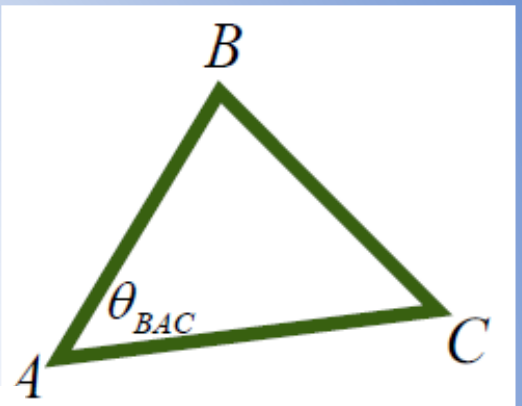
- The scalar component of **B** in the direction of the unit vector **a** is **B·a**
- The vector component of **B** in the direction of the unit vector **a** is **(B·a)a**

$$\mathbf{B}\cdot\mathbf{a} = |\mathbf{B}||\mathbf{a}|\cos\theta_{Ba} = |\mathbf{B}|\cos\theta_{Ba}$$

B.a is the projection of **B** in **a** direction.

Example

The three vertices of a triangle are located at $A(6,-1,2)$, $B(-2,3,-4)$, and $C(-3,1,5)$. Find: (*a*) $\mathbf{R}_{AB}$; (*b*) $\mathbf{R}_{AC}$; (*c*) the angle θ_{BAC} at vertex A; (*d*) the vector projection of $\mathbf{R}_{AB}$ on $\mathbf{R}_{AC}$.



$$\mathbf{R}_{AB} = (-2\mathbf{a}_x + 3\mathbf{a}_y - 4\mathbf{a}_z) - (6\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z) = \underline{-8\mathbf{a}_x + 4\mathbf{a}_y - 6\mathbf{a}_z}$$

$$\mathbf{R}_{AC} = (-3\mathbf{a}_x + 1\mathbf{a}_y + 5\mathbf{a}_z) - (6\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z) = \underline{-9\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z}$$

$$\mathbf{R}_{AB} \cdot \mathbf{R}_{AC} = |\mathbf{R}_{AB}||\mathbf{R}_{AC}| \cos\theta_{BAC}$$

$$\Rightarrow \cos\theta_{BAC} = \frac{\mathbf{R}_{AB} \cdot \mathbf{R}_{AC}}{|\mathbf{R}_{AB}||\mathbf{R}_{AC}|} = \frac{(-8\mathbf{a}_x + 4\mathbf{a}_y - 6\mathbf{a}_z) \cdot (-9\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z)}{\left|\sqrt{(-8)^2 + (4)^2 + (-6)^2}\right| \left|\sqrt{(-9)^2 + (2)^2 + (3)^2}\right|} = \frac{62}{\left|\sqrt{116}\right| \left|\sqrt{94}\right|} = 0.594$$

$$\Rightarrow \theta_{BAC} = \cos^{-1}(0.594) = \underline{53.56^\circ}$$

$$\mathbf{R}_{AB} \text{ on } \mathbf{R}_{AC} = (\mathbf{R}_{AB} \cdot \mathbf{a}_{AC})\mathbf{a}_{AC}$$

$$= \left((-8\mathbf{a}_x + 4\mathbf{a}_y - 6\mathbf{a}_z) \frac{(-9\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z)}{\left|\sqrt{(-9)^2 + (2)^2 + (3)^2}\right|} \right) \frac{(-9\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z)}{\left|\sqrt{(-9)^2 + (2)^2 + (3)^2}\right|}$$

$$= \frac{62}{\sqrt{94}} \frac{(-9\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z)}{\sqrt{94}}$$

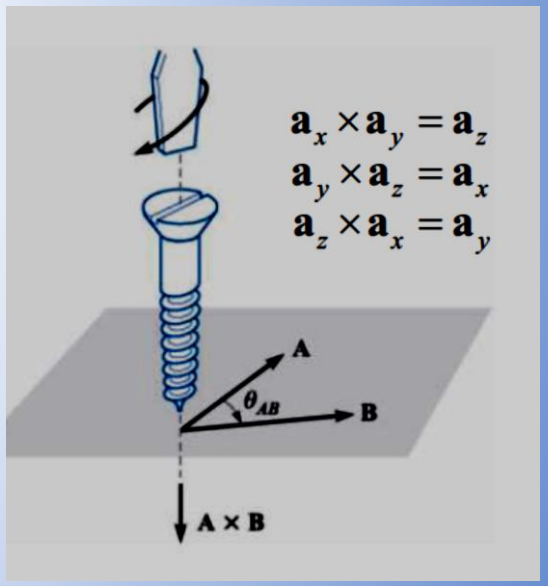
$$= \underline{-5.963\mathbf{a}_x + 1.319\mathbf{a}_y + 1.979\mathbf{a}_z}$$

The cross product

- Given two vectors **A** and **B**, the magnitude of the *cross product*, or *vector product*, written as **A**×**B**, is defined as the product of the magnitude of **A**, the magnitude of **B**, and the sine of the smaller angle between them.

- The direction of **A**×**B** is perpendicular to the plane containing **A** and **B** and is in the direction of advance of a right-handed screw as **A** is turned into **B**.

$$\mathbf{A}\times\mathbf{B} = \mathbf{a}_N |\mathbf{A}||\mathbf{B}| \sin\theta_{AB}$$



- The cross product is a vector, and it is not commutative:

$$(\mathbf{B}\times\mathbf{A}) = -(\mathbf{A}\times\mathbf{B})$$

The cross product

■ Example
Given $\mathbf{A} = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$ and $\mathbf{B} = -4\mathbf{a}_x - 2\mathbf{a}_y + 5\mathbf{a}_z$, find $\mathbf{A} \times \mathbf{B}$.

$$\mathbf{A}\times\mathbf{B} = (A_yB_z - A_zB_y)\mathbf{a}_x + (A_zB_x - A_xB_z)\mathbf{a}_y + (A_xB_y - A_yB_x)\mathbf{a}_z$$
$$= \big((-3)(5) - (1)(-2)\big)\mathbf{a}_x + \big((1)(-4) - (2)(5)\big)\mathbf{a}_y + \big((2)(-2) - (-3)(-4)\big)\mathbf{a}_z$$
$$= \underline{\underline{-13\mathbf{a}_x - 14\mathbf{a}_y - 16\mathbf{a}_z}}$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

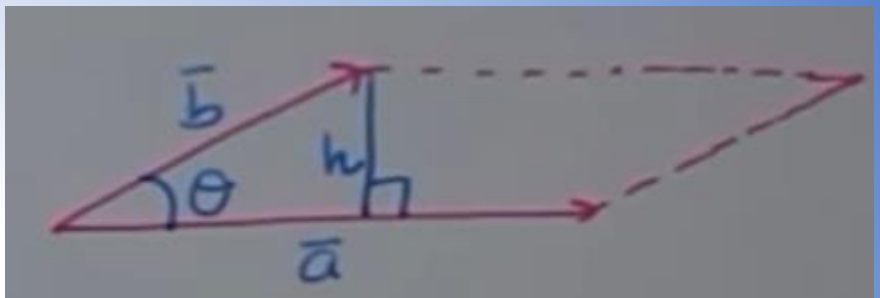
The three vertices of a triangle are located at $A(6,-1,2)$, $B(-2,3,-4)$, and $C(-3,1,5)$.

Find:

(a) $\mathbf{R}_{AB} \times \mathbf{R}_{AC}$;

(b)the area of the triangle;

(c) a unit vector perpendicular to the plane in which the triangle is located.



$$\text{Area}_{\triangle} = |\bar{a}|\,|\bar{b}| \sin\theta = |\bar{a} \times \bar{b}|$$

$$R_{AB} = -8\bar{a}_x + 4\bar{a}_y - 6a_z$$

$$R_{AC} = -9\bar{a}_x + 2\bar{a}_y + 3\bar{a}_z$$

$$\text{Area}_{\triangle} = \frac{1}{2}\left|R_{AB} \times R_{AC}\right| = 42$$

$$R_{AB} \times R_{AC} = \begin{vmatrix} a_x & a_y & a_z \\ -8 & 4 & -6 \\ -9 & 2 & 3 \end{vmatrix}$$

$$= 24\bar{a}_x + 78\bar{a}_y + 20a_z$$

$$a_N = \frac{R_{AB} \times R_{AC}}{\left|R_{AB} \times R_{AC}\right|}$$

$$= 0.286\bar{a}_x + 0.928\bar{a}_y + 0.238\bar{a}_z$$


Thank
you